Control metodos #2

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Curso: Metodos numericos

Fecha: 07/07/2026

10/10

C = 17

$f(x) = x(\text{Sen}(cx)) - e^x$

$h = 0.5$

$x = \frac{\pi}{4}$

$f_1 = (x_0 + h)$

$f_{-1} = (x_0 - h)$

$f(x_0) = \frac{f_1 - f_{-1}}{2h}$

$f_1 = f(x_0 + h) = f(\frac{\pi}{4} + 0.5) = 1.2854 = A(\text{Sen}(17A)) - e^A = -3.4376$

$f_{-1} = f(x_0 - h) = f(\frac{\pi}{4} - 0.5) = 0.2854 = B(\text{Sen}(17B)) - e^B = -1.6129$

$f'(x_0) \cong \frac{-3.4376 + 1.6129}{2 \times 0.5} \cong -1.8247 \rightarrow$ VAproximado

DCI

$f(x) = x(\text{Sen}(17x)) - e^x$

$f'(x) = \text{Sen}(17x) + x \cdot \cos(17x) \cdot 17 - e^x$

$f'(\frac{\pi}{4}) = \text{Sen}(17\frac{\pi}{4}) + \frac{\pi}{4}\cos(17\frac{\pi}{4}) \cdot 17 - e^{\frac{\pi}{4}} = 7.9549$ Valor real

$Ea = |V_{real} - V_{aproximado}|$

$Ea = |7.9549 - (-1.8247)|$

$Ea = 9.7796$

Escaneado con CamScanner